

A crossed random-effects latent class model for annotation ensembles of large language models

İhsan Tolga Medeni^{1*}

^{1*}Department of Management Information Systems, Faculty of Business, Ankara Yıldırım Beyazıt University, Ankara, Türkiye.

Corresponding author(s). E-mail(s): ihasantolgamedeni@aybu.edu.tr;

Abstract

See anonymised manuscript.

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ORCID: İhsan Tolga Medeni, 0000-0002-0642-7908.

Declarations

- **Funding:** The author received no specific funding for this work.
- **Competing interests:** The author declares no competing interests.
- **Ethics approval and consent:** Not applicable. The study uses simulated data and publicly available legislative texts with published human codes; no human participants were involved.
- **Data availability:** The Congressional bill titles and their human-coded major topics are from the Comparative Agendas Project / Congressional Bills Project (<https://www.comparativeagendas.net>, accessed 25 September 2026) and are redistributed under CC BY-NC-SA 4.0 with attribution. All LLM annotations generated for this study, the five prompt templates, model version identifiers and query dates, and the full per-replication simulation output are archived at Zenodo: <https://doi.org/10.5281/zenodo.22956083>.

- **Code availability:** R and Stan code reproducing all simulations, the application, figures and tables is available at <https://github.com/tolgamedeni/cre-lcm-adac> and archived at the Zenodo record above.
- **Author contributions:** Sole author: conceptualisation, methodology, software, analysis, writing.
- **Use of AI tools:** Large language models are the object of study in Section 5 and are documented there (model versions, prompts, dates). In addition, an AI coding assistant (Claude Code, Anthropic) was used to write and test parts of the R and Stan code and to assist with LaTeX formatting and copy-editing; all methodological decisions, results and text were reviewed and are the responsibility of the author.